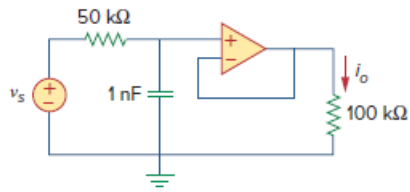


Problem

Compute $i_o(t)$ in the op amp circuit in Fig. 10.115 if $v_s = 4 \cos(10^4 t)$ V.

Figure 10.115:



Step-by-step solution

Step 1 of 4

$$4 \cos(10^4 t) \Rightarrow 4 \angle 0^\circ$$

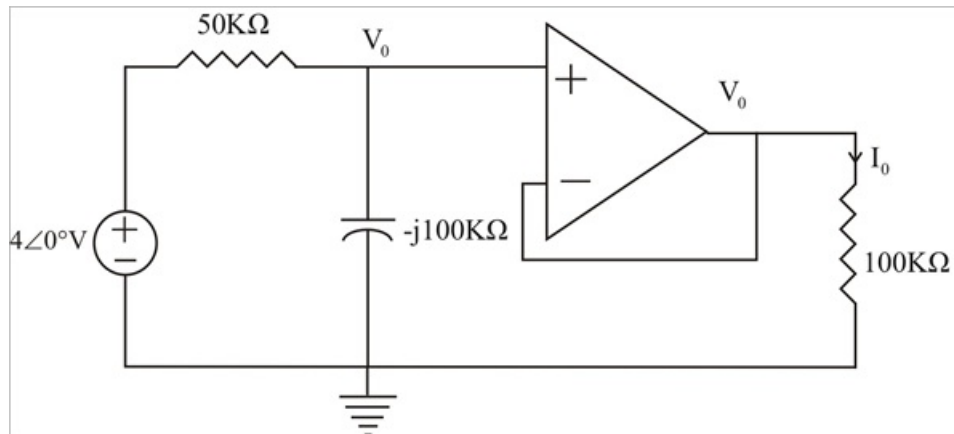
$$\omega = 10^4$$

$$1 \text{ nF} \Rightarrow \frac{1}{j\omega C}$$

$$\frac{1}{j(10^4)(10^{-9})} = -j100 \text{ k}\Omega$$

Step 2 of 4

Consider the circuit as shown below.



Step 3 of 4

At the non-inverting node,

$$\frac{4 - V_0}{50 \times 10^3} = \frac{V_0}{-j100 \times 10^3}$$

$$\frac{4 - V_0}{50} = \frac{V_0}{-j100}$$

$$4 - V_0 = \frac{V_0}{-j2}$$

$$V_0 = \frac{4}{1 + j0.5}$$

Step 4 of 4

Therefore,

$$I_0 = \frac{V_0}{100 \times 10^3}$$

$$I_0 = \frac{4}{100(1 + j0.5)} \text{ mA}$$

$$I_0 = 35.78 \angle -26.56^\circ \mu\text{A}$$

Therefore,

$$\therefore \boxed{i_0(t) = 35.78 \cos(10^4 t - 26.56^\circ) \mu\text{A}}$$